

Towards a MATSim Scenario for a Data-Scarce Megacity: Dhaka, Bangladesh

Adnan Abir^{1,*}, Grace O. Kagho², Tasnia Tabassum Prima³, Sk. Md. Mashrur¹

¹ Department of Civil Engineering, Bangladesh University of Engineering and Technology, Dhaka 1000, Bangladesh

² Institute of Construction and Infrastructure Management, ETH Zurich, Switzerland

³ Department of Computer Science and Engineering, Brac University, Dhaka 1212, Bangladesh

* Corresponding author. Email: 2004153@ce.buet.ac.bd Postal address: Department of Civil Engineering, Bangladesh University of Engineering and Technology, Dhaka 1000, Bangladesh

Presentation will be in person.

1. Introduction

Synthetic population and transport supply are the key requirements for Agent-based models such as MATSim (Horni et al. 2016). Many open pipelines have been developed for data-rich regions. For example, Eqasim (Hörl and Balać 2021) assumes weighted census microsamples, official building registries and standardized GTFS (General Transit Feed Specification). But most Global South megacities cannot supply these inputs as these inputs are missing or in a very aggregated unusable format. As a result, modelers either simplify the supply side or abandon network routing altogether. Due to the unavailability of reliable transit route data, the existing MATSim application for Dhaka, a mode-choice model for the planned Bus Rapid Transit corridor, relied on teleportation for public and non-motorized transport (Zannat et al. 2024).

Dhaka illustrates the problem in a compact form. Its finest-grained census exists only as an unstructured 1,668-page community report, its household travel survey (HTS) records no destination coordinates, and no GTFS feed exists for the city or for Bangladesh. The study area covers two city corporations and two adjoining upazilas, close to 13 million residents, with a modal split dominated by walking (32.8%) and rickshaw (29.7%), followed by car (16.3%), conventional bus (11.3%) and CNG auto-rickshaw paratransit (8.9%).

This study paves a path to the first MATSim model for Dhaka that simulates travel on the actual road with consideration of the public transport network. Two contributions may be useful for MATSim users: a population synthesis approach that uses iterative proportional updating (IPU) to adjust a travel survey using control totals from a printed census report, and a transit feed reconstructed from older operator route data to enable realistic public transport routing on the network.

2. Methodology

Five data sources have been used: the Dhaka Transport Coordination Authority (DTCA) HTS, the BBS (Bangladesh Bureau of Statistics) 2022 census community report, ward boundary shapefiles, a nationwide OpenStreetMap (OSM) building extract, and a self-built GTFS feed paired with an OSM road network (Figure 1). The synthesis process is an adapted version of the open-source Eqasim / synpp framework.

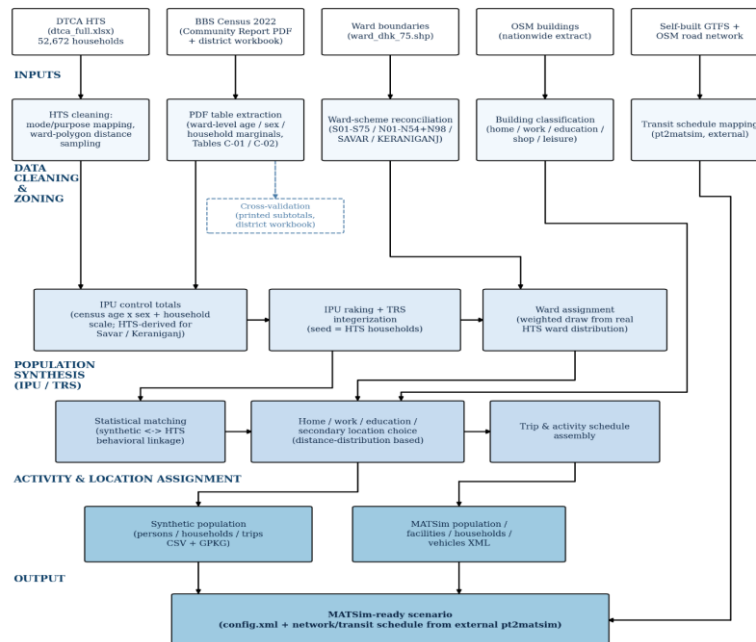


Figure 1. Data preparation pipeline for the Dhaka MATSim scenario.

2.1 Input preparation

The DTCA survey is a cross-sectional survey of 52,672 households and 186,988 members with 380,092 trip legs. It lists every household member directly. Restricting the study area leaves 38,092 households, 133,335 persons and 276,060 trips. Ward-level population, sex and age-group totals were parsed from the printed subtotal rows of the census PDF. Two independent tables agree on every ward total, and the ward sums reproduce the printed city-corporation subtotals. Trip purposes were collapsed to home, work, education and other. Because the two corporations use overlapping ward numbers, zone identifiers were prefixed by corporation (S07, N54), giving 130 wards plus two coarse peripheral zones.

2.2 Population synthesis

Individual-level census microdata are not available for Dhaka, so households are produced by reweighting the survey sample against external control totals rather than by expanding a microsample. The control totals combine two census sources at different spatial resolutions. Ward-level marginals for household counts, population by sex and population by age group come from the BBS community report. Household-size distributions, which the report does not tabulate at ward level, are taken from the district workbook and rescaled to local household counts. For Savar and Keraniganj, where ward-level census coverage does not exist, DTCA survey distributions serve as fallback totals.

Survey expansion factors initialize the weights, which are fitted by IPU (Ye et al. 2009; Kagho et al. 2020), an iterative heuristic that controls household- and person-level marginals simultaneously, to a tolerance of 10^{-3} or a maximum of 300 iterations. The continuous weights are integerized by truncate-replicate-sample. Each household is then assigned a ward by a weighted draw from the census ward distribution and linked to a survey record by statistical matching. At full expansion the 38,092 donor households scale to roughly 2.7 million synthetic households.

2.3 Activity locations

POI records from a Bangladesh OSM extract were used to identify work, education, commercial and leisure candidates, and untyped footprints within a floor-area range of 40–400 m². Home locations are drawn within the assigned ward. Work and education locations are matched by a k-d tree nearest-neighbour search against target distances drawn from the empirical HTS distance distributions, which substitutes for the commuting O-D matrix. The education assignment success rate reaches about 91%. Secondary locations use the relaxation-discretization approach (Hörl and Axhausen 2020) through the CARLA algorithm (Petre et al. 2025).

2.4 Transit supply

Legacy, non-machine-readable bus route inventories from DTCA were digitized into a GTFS feed. Because operations in the informal network follow frequency-based dispatching rather than published timetables, services were modelled with uniform headways of 10 minutes off-peak and 5 minutes in peak hours. Inter-stop travel times were obtained by querying OpenRouteService for sample stop pairs and calibrating free-flow speeds that were then extended across the network. The feed is mapped to the OSM network with pt2matsim (matsim-org 2016) to produce the transit schedule and vehicles.

3. Results

The synthetic population was validated against ward-level census controls. The age structure closely matches the census, with differences of less than 0.3 percentage points across all five-year age groups. The 0–4 age group is empty because the synthetic population is restricted to individuals aged five and above. The sex distribution is also well reproduced, with approximately 53% male and 47% female compared with the census shares of 54% and 46% (Figure 2a). Household size, controlled using the rescaled district-level distribution, differs from the census by less than 0.15 percentage points across all size categories.

Employment status and driving-license ownership were not used as control dimensions because no ward-level census benchmarks were available. The resulting shares are consistent with the survey, with around 33% of the population employed and 11% holding a driving license. This indicates that the statistical matching procedure transfers individual attributes without substantial distortion. The low level of license ownership is also consistent with the observed travel pattern. Walking, rickshaw, and paratransit account for roughly 70% of trips which is particularly relevant for subsequent mode-choice estimation.

The synthetic travel demand also reproduces the main characteristics of the observed activity patterns. Crow-fly distance distributions for home, work, education, and other activity locations follow the observed distributions, indicating that the distance-based location assignment can reproduce travel distances without relying on an origin–destination matrix. Education locations were successfully assigned for approximately 91% of students.

The reconstructed public transport supply includes 77 routes and 970 stops and is integrated with a road network containing 226,602 links (Figure 2b). As a result, public transport is routed directly on the network rather than being teleported. This represents the main practical improvement over the earlier Dhaka model.

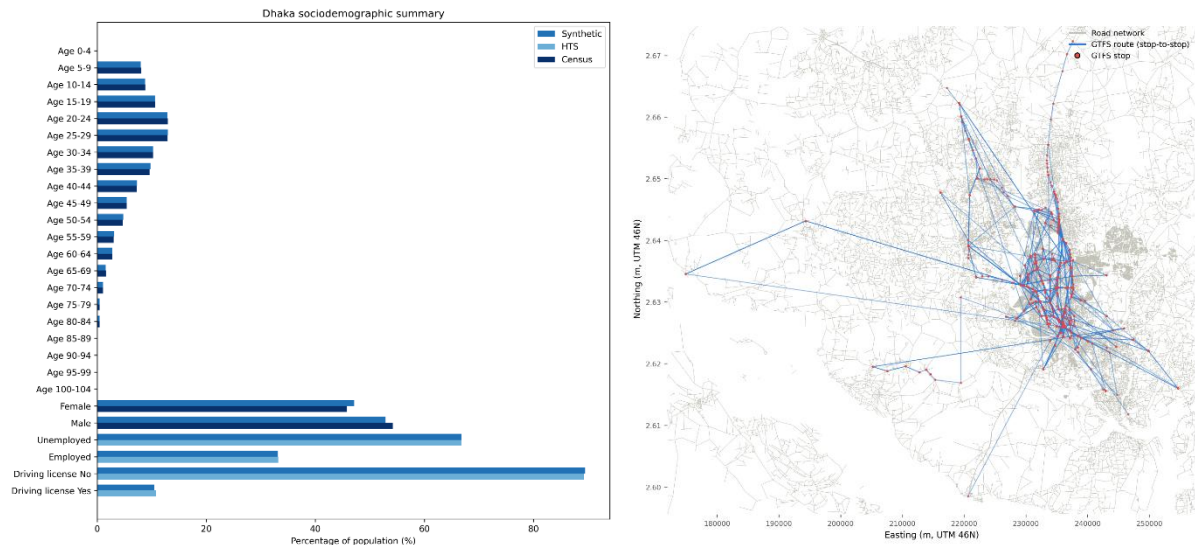


Figure 2. (a) Synthetic population against census controls and the survey for age, sex, employment and driving license; (b) reconstructed Dhaka Road network and public transport routes.

4. Conclusions and outlook

This study shows that a network-routed multimodal MATSim model can be assembled for a megacity that has neither a census microsample nor a transit feed, using a printed census report, a household travel survey and open geodata. The transferable steps are the recovery of control totals from unstructured official reports, IPU raking of the survey sample in place of microsample expansion, distance-distribution matching in place of commuting flow matrices, and frequency-based GTFS reconstruction.

Three main limitations should be noted. First, the peripheral upazilas rely mainly on survey expansion factors, which may not fully capture local population characteristics. Second, the diversity of household characteristics within each class is limited. Third, the input data predate the opening of MRT Line 6 in December 2022; therefore, the current scenario represents a pre-metro baseline rather than the present transport system. Ongoing work focuses on extending this baseline by incorporating MRT Line 6 to assess its effects on modal shift and network-wide congestion. A discrete mode-choice model is also being developed for the scoring function. The same population synthesis and demand-generation pipeline will further be tested in other South Asian cities to assess its transferability.

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